

Orthonormal Bases

$q_1, q_2, q_3, \dots, q_n$ vectors are orthonormal if

$$q_i^T q_j = \begin{cases} 0, & \text{when } i \neq j \\ 1, & \text{when } i = j \end{cases}$$

But here, what does orthonormal mean? Also what does the definition mean too?

↳ A basis is orthonormal if:

↳ Orthogonality → Every pair of vectors is perpendicular.
Meaning, their dot product is zero.

↳ Normalized → Each vector has length 1

So, a matrix Q with orthonormal columns satisfies $Q^T Q = I$:

$$Q^T Q = \begin{bmatrix} -q_1^T- \\ -q_2^T- \\ -q_3^T- \\ \vdots \\ -q_n^T- \end{bmatrix} \begin{bmatrix} | & | & | & \dots & | \\ q_1 & q_2 & q_3 & \dots & q_n \\ | & | & | & \dots & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} = I.$$

⊗ Here, $Q^T Q = I \Rightarrow Q^T = Q^{-1}$.
↳ Definition of orthogonal matrix: $A^T = A^{-1}$.

Example 1:

$$Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\leadsto Q Q^T = I$$

$$\hookrightarrow Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

⊗ Every permutation matrix is an orthogonal matrix ($P^T = P^{-1}$)
But why? (Actually they are orthonormal too)

/

... it doesn't

But Why!

→ A permutation matrix only re-arranges basis vectors, so it doesn't change their length or angles between them. Since permutations are just row swaps, applying P once shuffle rows, and applying P^T (swapping them back) restores the original arrangement.

Example 2:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \rightarrow \text{A rotation matrix.}$$

→ Here, $Q^T = Q^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

→ This is a orthogonal matrix because: $C_1 = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$ and $C_2 = \begin{bmatrix} \sin\theta \\ \cos\theta \end{bmatrix}$

$$C_1 \cdot C_2 = (\cos\theta)(-\sin\theta) + (\sin\theta)(\cos\theta) = 0$$

→ This is normalized because: $C_1^T C_1 = \begin{bmatrix} \cos\theta & \sin\theta \end{bmatrix} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} = \cos^2\theta + \sin^2\theta = 1$

$$C_2^T C_2 = \begin{bmatrix} -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \sin\theta \\ \cos\theta \end{bmatrix} = \sin^2\theta + \cos^2\theta = 1.$$

Reflection Matrix

→ $Q = I - 2uu^T$

Here, u is a unit vector

At first, let us

prove, $Q = I - 2uu^T$

Proof:

① Any vector v can be split into two components.

$$v = v_x + v_y$$

Since, v_x is the parallel part,

$$v_x = u \left(\frac{u^T v}{u^T u} \right)$$

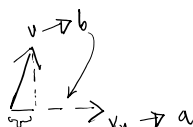
$v_x \rightarrow$ with respect to x -axis $\rightarrow$ parallel to u
 $v_y \rightarrow$ with respect to y -axis $\rightarrow$ perpendicular to u .

consider u as the unit vector of v

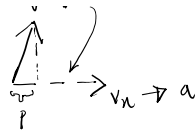
→ $u = \hat{v} = \frac{v}{\|v\|}$

→ Projection vector, $p = \hat{a}a = \frac{a^T b}{a^T a} a$.

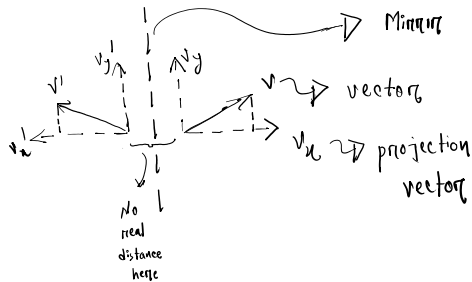
→ $v_x = u(u^T v)$ $\left[u^T u = 1 \text{ as } \|u\| = 1 \right]$



$$\Rightarrow v_x = u(u^T v) \quad \left[u^T u = 1 \text{ as } \|u\| = 1 \right]$$



ii) Now, let us visualize how does the reflection work?



From here, we can clearly see that both the length and direction stays the same for v_y .

$$v_y' = v_y$$

For, v_x it goes to the opposite direction.

$$v_x' = -v_x$$

iii) So, $v = v_x + v_y \quad \Rightarrow \quad v' = v_x' + v_y'$
 $= -v_x + v_y$

$$\Rightarrow v' = v_x + v_y - 2v_x$$

$$= v - 2v_x$$

$$= v - 2u(u^T v)$$

$$= (I - 2uu^T)v$$

$$v = Iv$$

$\Rightarrow$ multiplying any vector by the identity matrix I gives the same vector.

Since, $v' = (I - 2uu^T)v$ and this

holds for any v , so the

matrix representing the transformation: $Q = I - 2uu^T$

Another property for reflection matrix is that: $Q = Q^T = Q^{-1}$

And of course: $Q^T Q = I$, $Q^2 = I$ (Easy proof)

A reflection matrix is a self type of orthonormal matrix that reflects vector across a hyperplane.

For example:

i) Reflection across the x -axis $\rightarrow Q = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

ii) Reflection across the line $y=x \rightarrow Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

⊗ All reflection matrices are orthonormal but not all orthonormal matrices are matrices.

Same goes for rotations and permutations are matrices.

So, Rotations, Reflections, Permutations $\in$ Orthonormal.

Projections QQ^T using Orthonormal Bases: Q replaces A

We know that projection matrix, $P = A(A^T A)^{-1} A^T$

So, for orthonormal matrix, $P = Q(Q^T Q)^{-1} Q^T$
 $= QQ^T$

So, if Q is square, $P = I$

$Q^T Q = I$ (true for any dimension)

But

$QQ^T = I$ (true for only square matrix)

⊗ Now for orthonormal columns, $\hat{x} = Q^T b$ and $p = Q\hat{x}$ and $p = QQ^T b$

So, $Q^T Q \hat{x} = Q^T b$

$\hookrightarrow \hat{x} = Q^T b \quad \left(\hat{x}_i = q_i^T b \right)$

Gram-Schmidt Process

$\hookrightarrow$ a process of converting a set of linearly independent vectors into an orthonormal set of vectors.

Let's break it down using three vectors a, b, c :

Step 1: no constraint in the first vector.

$$A = a.$$

Step 2: Since b is not necessarily orthogonal to a , we remove its projection onto A :

$$B = b - \frac{A^T b}{A^T A} A.$$

Ensures B is perpendicular to A . Subtracting the projection removes any component in the direction of A .

Step 3: Since c might not be perpendicular to both A and B , we remove its projection along both directions:

$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B$$

Now, C is perpendicular to both A and B .

Step 4: Now, normalize the vectors:

$$q_1 = \frac{A}{\|A\|}, \quad q_2 = \frac{B}{\|B\|}, \quad q_3 = \frac{C}{\|C\|}$$

Factorization of $A = QR$

Let us consider a matrix A with vector a, b, c .

We will find orthogonal columns which is q_1, q_2, q_3 and it will be in matrix Q .

Then, $A = QR$.

$$\begin{bmatrix} | & | & | \\ a & b & c \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ & q_2^T b & q_2^T c \\ & & q_3^T c \end{bmatrix}$$

upper
triangular

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & -3 \\ 0 & -2 & 3 \end{bmatrix}$$

Soln:

(Treating A, B, C as Q_1, Q_2, Q_3)

$$Q_1 = a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$Q_2 = b - \frac{Q_1^T b}{Q_1^T Q_1} Q_1 = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\left| \begin{array}{l} Q_1^T b = 2 - 0 + 0 = 2 \\ Q_1^T Q_1 = 1 + 1 = 2 \end{array} \right.$$

$$= \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$Q_3 = C - \frac{Q_2^T C}{Q_2^T Q_2} Q_2 - \frac{Q_1^T C}{Q_1^T Q_1} Q_1$$

$$= \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} - \frac{-6}{6} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} - \frac{6}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$Q_2^T C = 3 - 3 - 6 = -6$$

$$Q_2^T Q_2 = 1 + 1 + 4 = 6$$

$$Q_1^T C = 3 + 3 - 0 = 6$$

$$Q_1^T Q_1 = 2$$

$$\text{Now, } q_1 = \frac{Q_1}{\|Q_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad q_2 = \frac{Q_2}{\|Q_2\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad q_3 = \frac{Q_3}{\|Q_3\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{2} & \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{6} & -\sqrt{6} \\ 0 & 0 & \sqrt{3} \end{bmatrix}$$

(cross check by $Q^{-1}A$)